Chapter 19

Medicines and Drugs

1. The Role of Medicines
2. Using Medicines Safely

Lesson 1 — The Role of Medicines

A **medicine** is a drug used to treat, prevent, or manage disease. A **drug** is any substance other than food that changes the structure or function of the body or mind. All medicines are drugs, but not all drugs are medicines.

The Opioid Crisis — Critical New Section

About **14% of high school students** report misusing prescription opioids. Overdose deaths involving fentanyl in 15–24 year-olds rose from **2.7% in 2018 to 80.8% in 2023**. Drug dealers now press fentanyl into **counterfeit pills** that look identical to real prescriptions — fake Xanax, fake Adderall, fake OxyContin.

> THE RULE IS ABSOLUTE: If a pill did not come from a licensed pharmacy with a valid prescription for YOU personally, do NOT take it. There is no visual way to tell a real pill from a fentanyl-laced counterfeit. One dose the size of a few grains of salt can be fatal.

Naloxone (Narcan) — now available OTC without a prescription — reverses opioid overdose when administered quickly. Knowing it exists and where to find it can save a life.

Lesson 2 — Using Medicines Safely

All new medicines in the U.S. must be reviewed and approved by the **FDA** before going to market — typically requiring laboratory testing, animal testing, and three or more phases of clinical trials in human volunteers.

Prescription medicines	Available only with written approval from a licensed prescriber; dispensed by a licensed pharmacist
OTC medicines	Available without a prescription; safe for self-directed use when label directions are followed
Medicine misuse	Using a medication in ways other than intended — even without intent to abuse
Medicine abuse	Intentionally using a medication for non-medical reasons; illegal and dangerous

1	**Read the entire label** Every time, before taking anything
2	**Never exceed recommended dose** Even OTC medications can cause severe damage in overdose
3	**Never share prescriptions** What helps one person can harm another — and it's illegal
4	**Tell your doctor everything** Every medication and supplement you take — interactions can be deadly
5	**Dispose of unused medications safely** Use pharmacy take-back programs; don't flush or leave accessible

Chapter Vocabulary

Lesson 1

Medicines	Drugs used to treat, prevent, or manage diseases or other medical conditions
Drugs	Substances other than food that change the structure or function of the body or mind
Vaccine	A preparation that trains the immune system to recognize and fight a specific disease without causing it
Side effects	Effects of a medication other than the intended therapeutic effect
Synergistic effect	When two medicines together produce a greater effect than either would produce alone

Lesson 2

Prescription medicines	Medications dispensed only with written approval from a licensed prescriber
Over-the-counter (OTC) medicines	Medications available without a prescription, approved for self-directed use
Medicine misuse	Using a medication in ways other than intended
Medicine abuse	Intentionally using a medication for non-medical reasons; illegal and dangerous
Drug overdose	A toxic, potentially fatal reaction when a drug is taken in an amount the body cannot safely process